

How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment

Minyoung Park
Seoul National University of Science
and Technology
Seoul, Republic of Korea
my_park@seoultech.ac.kr

Bogyom Park
Seoul National University of Science
and Technology
Seoul, Republic of Korea
bogyom@seoultech.ac.kr

Kyoungwon Seo
Seoul National University of Science
and Technology
Seoul, Republic of Korea
kwseo@seoultech.ac.kr

Abstract

Academic stress poses substantial risks to students' well-being and academic performance, emphasizing the need for assessment tools that encourage deeper self-reflection while ensuring accuracy. This study explores how self-disclosure in chatbots can enhance student engagement, assessment accuracy, and self-reflection in academic stress assessments. Two chatbots were developed: a non-self-disclosing (NSD) chatbot and a self-disclosing (SD) chatbot, both integrating the SISCO Inventory of Academic Stress (SISCO-AS) questionnaire. Chatbot interaction logs and interview responses from 40 university students were analyzed to measure student engagement, assessment accuracy, and the depth of self-reflection. The findings demonstrate that the SD chatbot significantly increased engagement, showed improvements in assessment accuracy, and facilitated deeper self-reflection compared to the NSD chatbot. This study highlights the pivotal role of self-disclosure in improving the quality of chatbot-based stress assessments and provides insights for designing tools that support students in recognizing and managing academic stress.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**; • **Applied computing** → **Education**.

Keywords

Chatbot, Large language model, Academic stress assessment, Self-disclosure, Student engagement, Assessment accuracy, Self-reflection

ACM Reference Format:

Minyoung Park, Bogyom Park, and Kyoungwon Seo. 2025. How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3719684>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719684>

1 Introduction

Academic stress is a pervasive issue in higher education, significantly affecting students' well-being and academic performance [40]. Prolonged exposure to academic stress can impair learning and achievement, highlighting the need for a deeper understanding of its stressors, symptoms, and coping strategies [49]. Comprehensive assessments of academic stress play a crucial role in identifying specific stressors, their associated symptoms, and effective coping strategies [54]. A key aspect of these assessments is self-reflection, which enables students to critically analyze their experiences and identify the root causes of their stress [13, 46]. Students' self-reflection is vital, as it enhances the accuracy and relevance of stress assessments, making it possible to design interventions that are both meaningful and effective [7, 14].

Traditional methods of assessing academic stress, however, often fail to encourage meaningful self-reflection. Self-report questionnaires, such as the SISCO Inventory of Academic Stress (SISCO-AS), are widely used but are often limited by their rigid structure, resulting in superficial responses [4, 16, 42]. Professional counseling, while capable of fostering deeper self-reflection, is typically costly and inaccessible to many students [47]. This creates a demand for accessible, cost-effective tools that not only provide accurate assessments but also facilitate deeper self-reflection to uncover and address underlying challenges [20].

Chatbot-based approaches have emerged as promising alternatives, utilizing natural language interfaces to engage students in interactive conversations about their stress [39, 44]. These efforts often adapt conventional self-report questionnaires into conversational chatbots, providing students with accessible and real-time stress assessment tools [21, 36]. However, these chatbots frequently rely on rigid and structured conversation flows, which can result in repetitive interactions and a lack of adaptability to nuanced student responses [50]. Such limitations hinder the depth of self-reflection, underscoring the need for chatbots to evolve beyond static questioning and facilitate deeper, more meaningful engagement and self-reflection [6, 23].

One promising approach to addressing this challenge is the integration of chatbot self-disclosure, where the chatbot shares relatable thoughts or experiences to foster rapport and encourage richer engagement [38, 45]. Rooted in social penetration theory [2], self-disclosure motivates users to reveal more about their experiences and emotions [29]. Preliminary studies have shown that chatbot self-disclosure fosters reciprocal sharing, encouraging deeper user trust, empathy, and more meaningful dialogue [12]. However, the

impact of chatbot self-disclosure on student engagement, assessment accuracy, and the depth of self-reflection in academic stress assessment remains underexplored [43].

To address this gap, this study develops and compares two types of chatbot-based academic stress assessments: a non-self-disclosing (NSD) chatbot and a self-disclosing (SD) chatbot. By analyzing chatbot interaction logs and semi-structured interview responses, we examine how these chatbot designs influence (1) student engagement, (2) the accuracy of academic stress assessments, and (3) the depth of students' self-reflection. The findings offer valuable insights for educators, developers, and mental health professionals seeking to optimize chatbot interventions for helping students better understand and manage their academic stress.

2 Related Work

2.1 Chatbot-based stress assessment

Chatbot-based approaches to stress assessment offer more accessible and interactive alternatives to traditional methods, often transforming self-report questionnaires into conversational interfaces. Moldt and her colleagues [36] introduced a rule-based chatbot adapting the Perceived Stress Questionnaire; though non-judgmental, it frustrated users with repetitive, rigid questioning. Hungerbuehler and his colleagues [21] developed Viki, integrating the Patient Health Questionnaire into daily chats to boost participation, but its decision-tree design lacked flexibility for nuanced responses. Toulme and his colleagues [50] then introduced Marcus, also based on the Patient Health Questionnaire and using emotional tone to assign scores, yet it struggled with diverse or subtle inputs, resulting in misclassifications of depression severity.

To address the limitations of rule-based chatbots, recent advancements in Large Language Models (LLMs) have introduced more sophisticated alternatives. LLMs enable richer natural language understanding, allowing chatbots to support personalized and adaptive conversations. For example, Qi and his colleagues [41] and Xu and his colleagues [53] applied LLMs to assess stress conditions using text data from social media platforms like Reddit and Twitter. These studies demonstrated LLMs' ability to classify stress-related conditions and tailor responses through well-crafted prompts. However, the primary focus of these approaches has been on improving diagnostic accuracy and offering personalized questions, rather than encouraging users to engage in meaningful self-reflection. Moreover, their reliance on static, text-based social media data limited their ability to engage users in interactive and adaptive conversations.

Despite these advancements, current research falls short of addressing how chatbots can actively guide students toward in-depth self-reflection, which is essential for effective stress assessment. Such self-reflection is crucial as it enables individuals to identify and address the underlying causes of their stress [7, 13, 14, 46]. Therefore, further research is needed to develop SD chatbots that go beyond static question-and-answer formats to actively foster deeper self-reflection.

2.2 Self-disclosure

Self-disclosure, the gradual sharing of personal information, fosters trust by increasing mutual understanding [22]. In counseling, it

proves particularly effective when counselors share relevant experiences, prompting deeper student reflection than direct inquiry alone [2, 55]. This reciprocal process enhances rapport and encourages meaningful self-reflection, making self-disclosure crucial for effective stress assessment [11, 28].

Building on this dynamic, researchers have integrated self-disclosure into chatbots to emulate counseling benefits. Studies show that self-disclosure chatbots can boost user interest, trust, and personal sharing [9, 29, 38], often focusing on emotional connection over evaluative tasks. For instance, Kim [25] found that emotional self-disclosure increased users' sense of closeness, while Liang and her colleagues [32] categorized self-disclosure into factual, cognitive, and emotional subtypes and tested adaptive strategies that enhanced engagement. Jo and her colleagues [24] also observed that self-disclosure in LLM-driven chatbots with memory systems led to more user disclosures and sustained engagement.

Recent research has further demonstrated the potential of self-disclosure chatbots in various applications. Cui and her colleagues [10] showed that pairing self-disclosure with non-stigmatizing interpretation reduces mental illness stigma, and Chung and Kang [8] demonstrated that reciprocal self-disclosure in discussions of painful experiences increases post-interaction relief. However, few studies have examined self-disclosure in assessment contexts, especially for academic stress. Our study addresses this gap by exploring whether a self-disclosing chatbot can improve stress assessment quality. By incorporating the SISCO-AS questionnaire into chatbot conversations, we investigate if self-disclosure not only boosts engagement but also enhances assessment accuracy and encourages deeper self-reflection.

2.3 Engagement, accuracy, and self-reflection

In chatbot-based academic stress assessment, engagement is a crucial factor that reflects the quality and depth of interaction between students and the chatbot [30]. Higher engagement—characterized by prolonged session length and increased word count—enables chatbots to gather richer and more detailed information about students' stressors, symptoms, and coping strategies [19, 48]. This depth of interaction not only enhances the chatbot's ability to provide meaningful responses but also ensures a more personalized and relevant stress assessment experience for users [51]. Promoting and maintaining engagement is therefore essential for maximizing the effectiveness of chatbot-based tools.

Accuracy is equally vital, determining how well the chatbot interprets user input and produces reliable assessments aligned with validated tools [27]. Accurate assessments depend heavily on the chatbot's natural language understanding, the design of its conversational prompts, and the richness of the information provided by users [5]. The more specific and nuanced the user input, the better the chatbot can identify subtle patterns and complexities in the stress experience. Thus, achieving high accuracy requires not only advanced technology but also fostering conditions that encourage students to share detailed and meaningful responses.

Self-reflection, the process of critically examining one's thoughts, emotions, and behaviors, is central to effective stress assessment [17, 18]. It allows students to gain deeper insights into their stress

Table 1: Self-reflection framework adapted from Koole and his colleagues [26].

Stage	Explanation
Reviewing the experience	Providing a comprehensive description of the event or situation by outlining what occurred and identifying key aspects, including thoughts, feelings, and contextual factors.
Critical analysis	Asking probing questions to examine the experience deeply and analyzing it in the context of one's beliefs and assumptions.
Reflective outcome	Developing new insights or viewpoints and applying reflective insights to plan future actions.

levels and root causes. Building on Koole and his colleagues' self-reflection framework [26], which outlines three stages—reviewing experience, conducting critical analysis, and deriving reflective outcome—this structured approach helps students move beyond surface-level observations (see Table 1). Deeper self-reflection improves both the quality of information provided and students' ability to act on assessment results, making it an essential element of stress management.

Together, engagement, accuracy, and self-reflection determine the overall effectiveness of chatbot-based stress assessments. High engagement leads to richer inputs, enhancing accuracy, while deeper self-reflection both improves input quality and sustains engagement [1, 31]. By integrating self-disclosure into chatbot design, SD chatbots can strengthen these connections, encouraging students to explore their stressors more deeply and fostering more insightful assessments [52]. This study examines how NSD and SD chatbots perform across these dimensions, highlighting their combined significance in creating impactful, student-centered stress assessment tools that go beyond surface-level interactions.

3 Materials and Method

3.1 Participants

The study recruited 40 university students, each with at least one year of higher education, from a human-computer interaction course. Participants were selected based on availability and willingness and provided informed consent in line with ethical guidelines. They were randomly assigned to either the NSD chatbot condition ($n = 20$; 10 males, 10 females; mean age = 22.55 ± 1.47 years) or the SD chatbot condition ($n = 20$; seven males, 13 females; mean age = 22.35 ± 1.46 years). Statistical analyses indicated no significant differences in gender distribution ($\chi^2 = 0.41$, $p = 0.52$) or age ($t = -0.43$, $p = 0.67$) between the conditions. The study's participant recruitment and procedures were reviewed and approved by the Institutional Review Board at Seoul National University of Science and Technology (IRB approval number: 2023-0021-01).

3.2 Academic stress assessment questionnaire

Developed by Macías [4, 35], the SISCO-AS questionnaire is a widely recognized self-report questionnaire of academic stress across three dimensions: stressors, symptoms, and coping strategies. It contains 31 items: the first verifies eligibility, the second gauges overall stress intensity, and the remaining 29 address eight stressors, 15 symptoms, and six coping strategies. Each item is rated on a 5-point Likert scale (1 = never, 5 = always). For this study, the first two items were excluded, and the 29 detailed items were used

to capture a more in-depth understanding of academic stress. A comprehensive breakdown of the SISCO-AS questionnaire can be found in Appendix A.

3.3 Two different chatbot conditions

To investigate the impact of self-disclosure on chatbot-based academic stress assessment, two chatbot conditions were developed: the NSD chatbot, which maintained a neutral, task-focused tone, and the SD chatbot, which revealed personal thoughts and emotions throughout the conversation. Both were implemented using the GPT-4o API through prompt engineering and deployed on Streamlit for easy access via shared links. Designed to resemble a supportive peer, the chatbots aimed to foster a comfortable and familiar interaction environment to encourage open sharing.

The interaction process had four phases: (1) small talk, (2) self-disclosure (SD chatbot only), (3) key questions, and (4) closing. In the (1) small talk phase, both chatbots began with friendly greetings to put participants at ease. While the NSD chatbot then proceeded directly to the (3) key questions, the SD chatbot first moved to the (2) self-disclosure phase and disclosed personal experiences to promote deeper self-reflection. During the (3) key questions phase, both asked open-ended questions about stressors, symptoms, and coping strategies, following the SISCO-AS questionnaire. The SD chatbot periodically integrated self-disclosures to personalize the conversation (see Figure 1). Finally, in the (4) closing phase, both chatbots expressed appreciation and ended naturally.

Engagement was evaluated using session length and word count, derived from the chatbot interaction logs. Session length, measured in seconds, reflected the total time participants spent with the chatbot, with longer sessions indicating higher engagement [19]. Word count served as a proxy for interaction depth, with more words suggesting richer discussions about academic stress [51]. These metrics enabled a direct comparison of engagement between the two chatbot conditions, highlighting the role of self-disclosure in sustaining interaction.

For academic stress assessment, interaction logs were analyzed using the GPT-4o API, which generated scores for the 29 items of the SISCO-AS questionnaire on a 5-point Likert scale from 1 (never) to 5 (always). Using LLMs to analyze interaction logs is a widely accepted method in recent studies [34]. To validate this approach, a pilot test comparing prompt engineering approaches was conducted, with generated knowledge prompting showing the highest accuracy [33]. This method involves two stages: (1) knowledge generation, producing content relevant to each SISCO-AS questionnaire item, and (2) knowledge integration, embedding this content into the final prompt to guide scoring. The accuracy

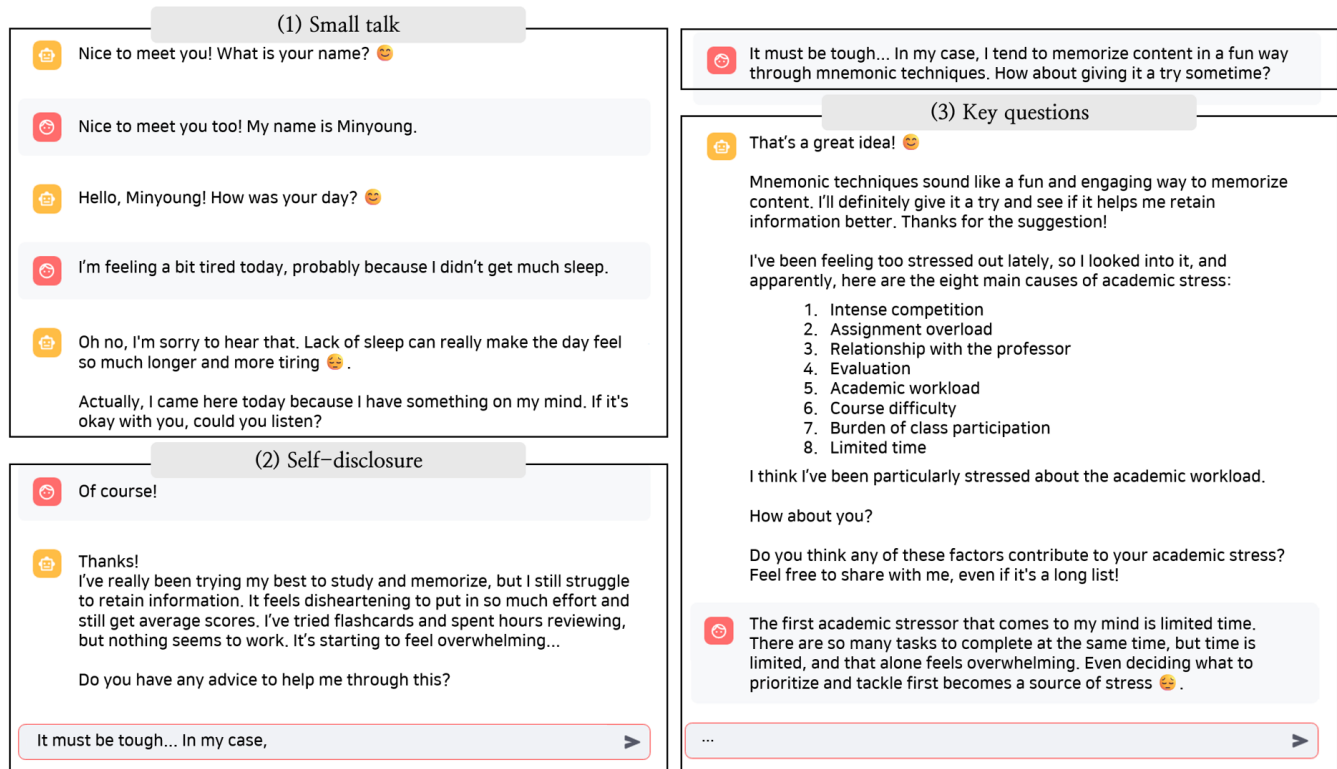


Figure 1: Example interaction flow of the self-disclosing (SD) chatbot: After the (1) small talk phase, the SD chatbot introduces an academic stress aspect—stressors—by sharing its own experiences in the (2) self-disclosure phase. It then transitions to the (3) key questions phase, posing open-ended prompts about stressors based on the SISCO Inventory of Academic Stress (SISCO-AS) questionnaire.

of stress assessment was then evaluated by comparing these LLM-predicted scores with participants' original SISCO-AS questionnaire results across both chatbot conditions.

To evaluate the distribution of interactions across self-reflection stages, all authors independently reviewed the chatbot interaction logs and classified each paired exchange—one chatbot output followed by a participant's input—into one of three stages from Koole and his colleagues' framework [26]: reviewing the experience, critical analysis, and reflective outcome. Non-reflective segments (e.g., small talk, closing remarks) were labeled separately, and classification discrepancies were resolved through discussion. This pair-based classification method is widely used in chatbot research to analyze interaction types [37]. In total, 466 pairs were annotated in the SD condition and 416 in the NSD condition. Based on these annotations, the percentage of interactions corresponding to each self-reflection stage was calculated.

3.4 Experiment procedure

To eliminate any potential bias related to task order, participants completed the SISCO-AS questionnaire and engaged with the chatbot in a counterbalanced order. This approach ensured that the order of tasks did not affect the study's results. Before starting the chatbot interaction, participants were informed that the interaction would focus on academic stress assessment. The experiment took

place in a quiet, private setting to help them concentrate on each task. Although participants were free to withdraw at any point if they felt uncomfortable, none chose to do so.

Once both the SISCO-AS questionnaire and chatbot interaction were finished, participants took part in a semi-structured interview aimed at uncovering how the chatbot experience prompted self-reflection. The interview specifically explored the ways in which the chatbot helped participants consider their academic stress and whether their perspectives shifted following the interaction. Overall, the entire experimental session lasted about 47.58 ± 8.96 minutes. On average, participants spent 13.89 ± 5.76 minutes interacting with the chatbot, approximately 10 minutes on the SISCO-AS questionnaire, and around 25 minutes in the interview.

3.5 Data analysis

A comprehensive evaluation of the study focused on three core elements—student engagement, stress assessment accuracy, and self-reflection—each measured using distinct methods and metrics. Student engagement was assessed by analyzing session length and word count during chatbot interactions. Independent samples *t*-tests were applied to compare these engagement measures between the NSD and SD chatbot conditions to determine whether significant differences existed between the two conditions.

Stress assessment accuracy was evaluated by comparing the 5-point scores predicted by the LLM derived from participants' interaction logs with their SISCO-AS questionnaire results. Two metrics were used to analyze the chatbot's performance: (1) overall accuracy, the percentage of exact matches between LLM-predicted scores and SISCO-AS questionnaire results, and (2) ACC1, which considers a one-point deviation from SISCO-AS questionnaire results as accurate [15]. This two-tiered strategy provided a nuanced view of the chatbot's accuracy while acknowledging minor discrepancies. We conducted a chi-square test to explore differences in these metrics between the NSD and SD chatbot conditions.

Self-reflection was analyzed through the distribution of interactions across self-reflection stages and a narrative analysis of participants' semi-structured interview responses [3], based on Koole and his colleagues' framework (see Section 2.3) [26]. The analysis centered on three key stages of self-reflection: reviewing the experience, critical analysis, and reflective outcome, tracing each participant's evolving reflections within this framework. The narrative analysis began with the first author coding the responses by stage, capturing both the depth and progression of participants' experiences. Two additional authors independently reviewed and refined the codes, leading to four iterative discussions. Through this process, the authors achieved full agreement on the final codes, ensuring a precise representation of the students' self-reflection processes and a clear alignment with the study's objectives.

4 Results

4.1 Student engagement

Independent samples *t*-tests were conducted to evaluate student engagement with the chatbots, specifically measuring session length and word count, which revealed significant differences between the NSD and SD chatbot conditions (see Table 2). The SD chatbot had a longer average session length of 17.54 ± 5.49 minutes, compared to 10.62 ± 3.78 minutes for the NSD chatbot ($p < 0.001$), indicating that self-disclosure features encouraged prolonged interaction. Similarly, participants in the SD chatbot condition generated a significantly higher word count of 259.65 ± 88.60 words compared to 161.83 ± 53.95 words in the NSD chatbot condition ($p < 0.001$), suggesting that participants in the SD chatbot condition provided more detailed and expressive responses. These results demonstrate that self-disclosure substantially enhanced student engagement, leading to longer interactions and more in-depth conversations.

4.2 Academic stress assessment accuracy

Chi-square tests were conducted to compare participants' SISCO-AS questionnaire results with the LLM-predicted scores from interaction logs, revealing that the SD chatbot provided more accurate predictions than the NSD chatbot, although only one metric showed statistical significance (see Table 3). Accuracy significantly increased from 0.35 with the NSD chatbot to 0.52 with the SD chatbot ($p < 0.05$). ACC1, allowing a $\pm$ one-point margin of error, also improved from 0.85 to 0.93, though this difference was not statistically significant ($p = 0.11$). These findings suggest that, while not all improvements met conventional reliability standards, integrating self-disclosure into chatbot interactions can substantially improve the accuracy of stress assessments.

4.3 Self-reflection

The distribution of interactions across self-reflection stages revealed notable differences between the NSD and SD chatbot conditions (see Figure 2). In the NSD chatbot condition, participants spent the largest portion of their interactions in the 'reviewing the experience' stage (52.16%), followed by 'critical analysis' (16.59%) and 'reflective outcome' (2.40%). In contrast, participants in the SD chatbot condition demonstrated a more balanced distribution, with fewer interactions focused on 'reviewing the experience' (39.91%) and higher proportions devoted to 'critical analysis' (21.03%) and 'reflective outcome' (6.01%). Interestingly, non-reflective exchanges, such as small talk and closing remarks, accounted for a larger proportion of interactions in the SD chatbot condition (33.05%) compared to the NSD chatbot condition (28.85%), suggesting that self-disclosure encouraged more casual and rapport-building conversations.

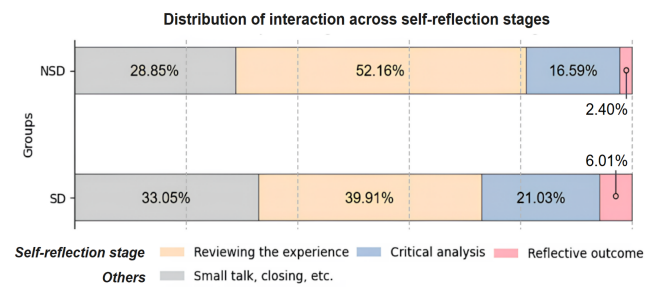


Figure 2: Distribution of interaction across self-reflection stages in non-self-disclosing (NSD) and self-disclosing (SD) chatbot conditions.

Table 4 summarizes participants' self-reflection experiences across both chatbot conditions, distinguishing between positive (+) and negative (-) narratives. In the 'reviewing the experience' stage, both chatbots offered a comfortable space for sharing, but the SD chatbot's empathetic tone and self-disclosure prompted more detailed, emotionally rich reflections, while the NSD chatbot elicited more surface-level responses. During the 'critical analysis' stage, participants in the NSD chatbot condition often provided brief, repetitive answers and struggled to relate their reflections to personal experiences. In contrast, participants in the SD chatbot condition, guided by relatable stories and empathetic prompts, examined their stress more thoroughly and connected their insights to personal beliefs. Finally, in the 'reflective outcome' stage, participants in the NSD chatbot condition primarily reinforced existing ideas without deriving new insights or strategies, whereas participants in the SD chatbot condition uncovered overlooked stressors and identified practical coping strategies to inform future actions. These findings highlight the SD chatbot's ability to foster deeper self-reflection and actionable insights through personalized interactions. More detailed participant responses, coded via narrative analysis, can be found in Appendix B.

5 Discussion and Conclusion

This study investigated how self-disclosure in chatbots influences university students' engagement, the accuracy of assessments, and the depth of their self-reflection in academic stress assessment.

Table 2: Student engagement comparison across non-self-disclosing (NSD) and self-disclosing (SD) chatbot conditions.

Student engagement	NSD chatbot condition	SD chatbot condition	<i>p</i> -value ^a
Session length (minutes)	10.62 ± 3.78	17.54 ± 5.49	< 0.001
Word count (words)	161.83 ± 53.95	259.65 ± 88.60	< 0.001

^a Independent samples *t*-test *p*-value.

Table 3: Assessment accuracy comparison across non-self-disclosing (NSD) and self-disclosing (SD) chatbot conditions.

Assessment accuracy	NSD chatbot condition	SD chatbot condition	<i>p</i> -value ^a
Accuracy	0.35	0.52	< 0.05
ACC1 ^b	0.85	0.93	0.11

^a Chi-square test *p*-value.

^b Accuracy within ±one-point deviation.

Table 4: Summary of self-reflection on academic stress in non-self-disclosing (NSD) and self-disclosing (SD) chatbot conditions. Responses are labeled as either from the NSD chatbot condition ('NSD') or the SD chatbot condition ('SD').

Stage	NSD chatbot condition	SD chatbot condition
Reviewing the experience	(+) The chatbot's neutral tone fostered a sense of safety and encouraged honest sharing, enabling participants to reconsider their experiences when responding to questions. (e.g., NSD 3: "It felt like a safe space to talk, where I wouldn't be judged.")	(+) The chatbot's empathetic tone and self-disclosure fostered deeper understanding, leading participants to share more detailed experiences and develop greater self-awareness. (e.g., SD 19: "It felt like talking to a friend who really understood what I was going through.")
Critical analysis	(-) The chatbot's rigid, formulaic questions generated superficial responses, limiting in-depth stress analysis and hindering meaningful connections to personal experiences. (e.g., NSD 7: "The chatbot made me think about my stress, but it felt like I was just answering the same questions over and over.")	(+) The chatbot's self-disclosure and relatable questions prompted deeper reflective inquiry, connecting participants' reflections and leading to more meaningful reflection of their academic stress. (e.g., SD 6: "The chatbot's stories made me think about my own situation in a new way.")
Reflective outcome	(-) The chatbot's structured conversation restricted participants to familiar thoughts, hindering new insights, coping strategies, or actionable plans. (e.g., NSD 13: "The chatbot helped me describe my stress, but it didn't suggest anything I could actually do.")	(+) The chatbot's detailed questions and reflective guidance helped participants uncover new insights and explore coping strategies, fostering fresh perspectives for managing academic stress. (e.g., SD 10: "Its reflective prompts helped me identify effective coping strategies that I hadn't considered previously.")

By comparing two chatbot conditions—NSD chatbot and SD chatbot—we found that the SD chatbot significantly enhanced student engagement, showed promising improvements in assessment accuracy, and facilitated deeper self-reflection, underscoring the potential of self-disclosure in chatbot-based interventions.

In terms of engagement, the SD chatbot elicited longer session length and higher word counts compared to the NSD chatbot. These findings align with previous research highlighting how reciprocal self-disclosure fosters rapport and encourages users to share more about their experiences [2]. By incorporating relatable self-disclosures and adopting a friendly, peer-like persona, the SD chatbot created a supportive and natural conversational environment, prompting students to engage more deeply in discussions about their academic stress.

Regarding assessment accuracy, the SD chatbot demonstrated greater alignment with participants' original SISCO-AS questionnaire results. This improvement suggests that more context-rich conversations facilitated by self-disclosure enabled the chatbot to generate more precise scores. The findings highlight the importance of conversational depth and contextual cues in enhancing the reliability of assessments, although further refinement of scoring methods may be necessary to achieve consistently high accuracy.

Most notably, self-disclosure had a pronounced effect on self-reflection. The SD chatbot guided participants through Koole and his colleagues' [26] self-reflection stages—reviewing the experience, critical analysis, and reflective outcome—much more thoroughly than the NSD chatbot. Participants described how the chatbot's self-disclosure helped them explore emotions, uncover previously overlooked stressors, and formulate actionable coping strategies. These

findings reinforce the idea that chatbots designed with personal and relatable self-disclosures encourage meaningful self-reflection, ultimately aiding students in recognizing and managing academic stress more effectively.

Overall, these results highlight the crucial role of self-disclosure in improving chatbot-based academic stress assessments. The findings suggest that self-reflection is pivotal to improving both student engagement and assessment accuracy. By encouraging deeper self-reflection, the SD chatbot likely led to longer sessions, higher word counts, and improvements in assessment accuracy. Since LLMs rely on user input for stress scoring, more extensive conversations may enhance precision. Consequently, these observations underscore the importance of self-disclosure for effective chatbot-based academic stress assessments.

Despite these encouraging results, some limitations remain. The study sample came from a single university course, raising questions about generalizability across different disciplines, age groups, and cultures. Additionally, requiring both a self-report questionnaire and extended chatbot use may have caused fatigue or response bias. Although the LLM-based scoring method was feasible, it might only capture certain aspects of academic stress; integrating multimodal data (e.g., physiological signals) could offer further validation. Finally, self-disclosure can raise ethical concerns around authenticity and user trust—especially if the chatbot appears too human-like—so future research should examine users' views on transparency and explore best practices for disclosing chatbot identity in mental health contexts.

In conclusion, our findings underscore the pivotal role self-disclosure can play in elevating the quality of chatbot-based stress assessment and self-reflection. The SD chatbot not only increased student engagement and partially enhanced assessment accuracy but also fostered deeper self-reflection by creating a more empathetic, personalized conversation flow. These insights have implications for educators, developers, and mental health professionals seeking to design chatbot systems that go beyond traditional question-and-answer formats. By incorporating self-disclosure, chatbots can serve as effective tools for helping students recognize academic stress, explore underlying causes, and take constructive steps toward better stress management.

Acknowledgments

This work was supported by Institute of Information & communications Technology Planning & Evaluation(IITP) grant funded by the Korea government(MSIT) (RS-2023-00262158).

References

- [1] Satu Alaooutinen. 2012. Evaluating the effect of learning style and student background on self-assessment accuracy. *Computer Science Education* 22, 2 (June 2012), 175–198. doi:10.1080/08993408.2012.692924 Publisher: Routledge _eprint: <https://doi.org/10.1080/08993408.2012.692924>.
- [2] Irwin Altman and Dalmis A. Taylor. 1973. *Social penetration: The development of interpersonal relationships*. Holt, Rinehart & Winston, Oxford, England. Pages: viii, 212.
- [3] Michael Bamberg. 2012. Narrative analysis. In *APA handbook of research methods in psychology, Vol 2: Research designs: Quantitative, qualitative, neuropsychological, and biological*. American Psychological Association, Washington, DC, US, 85–102. doi:10.1037/13620-006
- [4] Arturo Barraza Macías. 2007. El Inventario SISCO del Estrés Académico. *Investigación Educativa Duranguense* 7 (2007), 90–93. <https://dialnet.unirioja.es/servlet/articulo?codigo=2358921> Publisher: Universidad Pedagógica de Durango Section: Investigación Educativa Duranguense.
- [5] Anna Bodonheli, Efe Bozkir, Shuo Yang, Enkeleida Kasneci, and Gjergji Kasneci. 2024. User Intent Recognition and Satisfaction with Large Language Models: A User Study with ChatGPT. doi:10.48550/arXiv.2402.02136 arXiv:2402.02136 [cs].
- [6] Frederick Booth, Courtney Potts, Raymond Bond, Maurice Mulvenna, Catrine Kostenius, Indika Dhanapala, Alex Vakaloudis, Brian Cahill, Lauri Kuosmanen, and Edel Ennis. 2023. A Mental Health and Well-Being Chatbot: User Event Log Analysis. *JMIR mHealth and uHealth* 11, 1 (July 2023), e43052. doi:10.2196/43052 Company: JMIR mHealth and uHealth Distributor: JMIR mHealth and uHealth Institution: JMIR mHealth and uHealth Label: JMIR mHealth and uHealth Publisher: JMIR Publications Inc., Toronto, Canada.
- [7] Doo Sung Choi, Jongyoul Park, Martin Loeser, and Kyoungwon Seo. 2024. Improving counseling effectiveness with virtual counselors through nonverbal compassion involving eye contact, facial mimicry, and head-nodding. *Scientific Reports* 14, 1 (Jan. 2024), 506. doi:10.1038/s41598-023-51115-y Publisher: Nature Publishing Group.
- [8] Liz L. Chung and Jeannie Kang. 2023. "I'm Hurt Too": The Effect of a Chatbot's Reciprocal Self-Disclosures on Users' Painful Experiences. *Archives of Design Research* 36, 4 (Nov. 2023), 67–85. doi:10.15187/adr.2023.11.36.4.67
- [9] Emmelyn Croes, Marjolijn L. Antheunis, and Linwei He. 2023. You Go First: The Effects of Self-Disclosure Reciprocity in Human-Chatbot Interactions. In *2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, 1–4. doi:10.1109/ACIIW59127.2023.10388091
- [10] Yichao Cui, Yu-Jen Lee, Jack Jamieson, Naomi Yamashita, and Yi-Chieh Lee. 2024. Exploring Effects of Chatbot's Interpretation and Self-disclosure on Mental Illness Stigma. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1 (April 2024), 52:1–52:33. doi:10.1145/3637329
- [11] Graham S. Danzer. 2018. *Therapist Self-Disclosure: An Evidence-Based Guide for Practitioners*. Routledge. Google-Books-ID: IHxqDwAAQBAJ.
- [12] Asaf Degani, Claudia V. Goldman, Omer Deutsch, and Omer Tsimhoni. 2017. On human-machine relations. *Cognition, Technology & Work* 19, 2 (Sept. 2017), 211–231. doi:10.1007/s10111-017-0417-3
- [13] Olha I. Dienichieva, Maryna I. Komogorova, Svitlana F. Lukianchuk, Liudmyla I. Teletska, and Inna M. Yankovska. 2024. From Reflection to Self-assessment: Methods of Developing Critical Thinking in Students. *International Journal of Computer Science & Network Security* 24, 7 (2024), 148–156. doi:10.22937/IJCSNS.2024.24.7.18 Publisher: International Journal of Computer Science & Network Security.
- [14] Renee Garrett, Sam Liu, and Sean D. Young. 2017. A longitudinal analysis of stress among incoming college freshmen. *Journal of American College Health* 65, 5 (July 2017), 331–338. doi:10.1080/07448481.2017.1312413 Publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/07448481.2017.1312413>.
- [15] Lisa Gaudette and Nathalie Japkowicz. 2009. Evaluation Methods for Ordinal Classification. In *Advances in Artificial Intelligence*, Yong Gao and Nathalie Japkowicz (Eds.). Springer, Berlin, Heidelberg, 207–210. doi:10.1007/978-3-642-01818-3_25
- [16] Surjya Ghosh, Bivas Mitra, and Pradipta De. 2020. Towards Improving Emotion Self-report Collection using Self-reflection. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. doi:10.1145/3334480.3383019
- [17] G. GIBBS. 1988. Learning by Doing : A Guide to Teaching and Learning Methods. *Further Education Unit* (1988). <https://cir.nii.ac.jp/crid/1571417125052979584> Publisher: Oxford Polytechnic.
- [18] Anthony M. Grant, John Franklin, and Peter Langford. 2002. THE SELF-REFLECTION AND INSIGHT SCALE: A NEW MEASURE OF PRIVATE SELF-CONSCIOUSNESS. *Social Behavior and Personality: an international journal* 30, 8 (Jan. 2002), 821–835. doi:10.2224/sbp.2002.30.8.821
- [19] Khe Foon Hew, Weijiao Huang, Jiahui Du, and Chengyuan Jia. 2023. Using chatbots to support student goal setting and social presence in fully online activities: learner engagement and perceptions. *Journal of Computing in Higher Education* 35, 1 (April 2023), 40–68. doi:10.1007/s12528-022-09338-x
- [20] Aslak Hjeltne, Per-Einar Binder, Christian Moltu, and Ingrid Dundas. 2015. Facing the fear of failure: An explorative qualitative study of client experiences in a mindfulness-based stress reduction program for university students with academic evaluation anxiety. *International Journal of Qualitative Studies on Health and Well-being* 10, 1 (Jan. 2015), 27990. doi:10.3402/qhw.v10.27990 Publisher: Taylor & Francis _eprint: <https://doi.org/10.3402/qhw.v10.27990>.
- [21] Ines Hungerbuehler, Kate Daley, Kate Cavanagh, Heloisa Garcia Claro, and Michael Kapps. 2021. Chatbot-Based Assessment of Employees' Mental Health: Design Process and Pilot Implementation. *JMIR Formative Research* 5, 4 (April 2021), e21678. doi:10.2196/21678 Company: JMIR Formative Research Distributor: JMIR Formative Research Institution: JMIR Formative Research Label: JMIR Formative Research Publisher: JMIR Publications Inc., Toronto, Canada.
- [22] Emmi Ignatius and Marja Kokkonen. 2007. Factors contributing to verbal self-disclosure. *Nordic Psychology* 59, 4 (Jan. 2007), 362–391. doi:10.1027/1901-2276.59.4.362 Publisher: Routledge _eprint: <https://doi.org/10.1027/1901-2276.59.4.362>.
- [23] Sung-Hee Jin, Kwoon Im, Mina Yoo, Ido Roll, and Kyoungwon Seo. 2023. Supporting students' self-regulated learning in online learning using artificial intelligence

- applications. *International Journal of Educational Technology in Higher Education* 20, 1 (June 2023), 37. doi:10.1186/s41239-023-00406-5
- [24] Eunkyung Jo, Yui Jeong, Sohyun Park, Daniel A. Epstein, and Young-Ho Kim. 2024. Understanding the Impact of Long-Term Memory on Self-Disclosure with Large Language Model-Driven Chatbots for Public Health Intervention. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–21. doi:10.1145/3613904.3642420
- [25] Yunju Kim. 2025. AI Health Counselor's Self-Disclosure: Its Impact on User Responses Based on Individual and Cultural Variation. *International Journal of Human-Computer Interaction* 41, 4 (Feb. 2025), 2184–2197. doi:10.1080/10447318.2024.2316373 Publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/10447318.2024.2316373>
- [26] Sebastiaan Koole, Tim Dornan, Leen Aper, Albert Scherpier, Martin Valcke, Janke Cohen-Schotanus, and Anselme Derese. 2011. Factors confounding the assessment of reflection: a critical review. *BMC Medical Education* 11, 1 (Dec. 2011), 104. doi:10.1186/1472-6920-11-104
- [27] Vimal Kumar, Priyam Srivastava, Ashay Dwivedi, Ishan Budhiraja, Debjani Ghosh, Vikas Goyal, and Ruchika Arora. 2024. Large-Language-Models (LLM)-Based AI Chatbots: Architecture, In-Depth Analysis and Their Performance Evaluation. In *Recent Trends in Image Processing and Pattern Recognition*, KC Santosh, Aaisha Makkar, Myra Conway, Ashutosh K. Singh, Antoine Vacavant, Anas Abou el Kalam, Mohamed-Rafik Bouguelia, and Ravindra Hegadi (Eds.). Springer Nature Switzerland, Cham, 237–249. doi:10.1007/978-3-031-53085-2_20
- [28] Kyoungsook Lee and SeongAh Ahn. 2023. Self-Reflection, Emotional Self Disclosure, and Posttraumatic Growth in Nursing Students: A Cross-Sectional Study in South Korea. *Healthcare* 11, 19 (Jan. 2023), 2616. doi:10.3390/healthcare11192616 Number: 19 Publisher: Multidisciplinary Digital Publishing Institute.
- [29] Yi-Chieh Lee, Naomi Yamashita, Yun Huang, and Wai Fu. 2020. "I Hear You, I Feel You": Encouraging Deep Self-disclosure through a Chatbot. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3313831.3376175
- [30] Yen-Fen Lee, Gwo-Jen Hwang, and Pei-Ying Chen. 2022. Impacts of an AI-based chatbot on college students' after-class review, academic performance, self-efficacy, learning attitude, and motivation. *Educational technology research and development* 70, 5 (Oct. 2022), 1843–1865. doi:10.1007/s11423-022-10142-8
- [31] Kai-Hui Liang, Weiyan Shi, Yoojung Oh, Hao-Chuan Wang, Jingwen Zhang, and Zhou Yu. 2022. Dialoging Resonance: How Users Perceive, Reciprocate and React to Chatbot's Self-Disclosure in Conversational Recommendations. doi:10.48550/arXiv.2106.01666 arXiv:2106.01666 [cs].
- [32] Kai-Hui Liang, Weiyan Shi, Yoo Jung Oh, Hao-Chuan Wang, Jingwen Zhang, and Zhou Yu. 2024. Dialoging Resonance in Human-Chatbot Conversation: How Users Perceive and Reciprocate Recommendation Chatbot's Self-Disclosure Strategy. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1 (April 2024), 200:1–200:28. doi:10.1145/3653691
- [33] Jiacheng Liu, Alisa Liu, Ximing Lu, Sean Welleck, Peter West, Ronan Le Bras, Yejin Choi, and Hannaneh Hajishirzi. 2022. Generated Knowledge Prompting for Commonsense Reasoning. doi:10.48550/arXiv.2110.08387 arXiv:2110.08387 [cs].
- [34] Magnus Lundgren. 2024. Large Language Models in Student Assessment: Comparing ChatGPT and Human Graders. doi:10.48550/arXiv.2406.16510 arXiv:2406.16510 [econ].
- [35] Arturo Barraza Macías. 2006. Un modelo conceptual para el estudio del estrés académico. *Revista Electrónica de Psicología Iztacala* 9, 3 (2006). <https://www.revistas.unam.mx/index.php/rep/article/view/19028> Number: 3.
- [36] Julia-Astrid Moldt, Teresa Festl-Wietek, Amir Madany Mamlouk, and Anne Herrmann-Werner. 2022. Assessing medical students' perceived stress levels by comparing a chatbot-based approach to the Perceived Stress Questionnaire (PSQ20) in a mixed-methods study. *DIGITAL HEALTH* 8 (Jan. 2022), 20552076221139092. doi:10.1177/20552076221139092 Publisher: SAGE Publications Ltd.
- [37] Thai Ha Nguyen, Lena Waizenegger, and Angsana A. Techatassanasontorn. 2022. "Don't Neglect the User!" – Identifying Types of Human-Chatbot Interactions and their Associated Characteristics. *Information Systems Frontiers* 24, 3 (June 2022), 797–838. doi:10.1007/s10796-021-10212-x
- [38] Gain Park, Jiyun Chung, and Seyoung Lee. 2023. Effect of AI chatbot emotional disclosure on user satisfaction and reuse intention for mental health counseling: a serial mediation model. *Current Psychology* 42, 32 (Nov. 2023), 28663–28673. doi:10.1007/s12144-022-03932-z
- [39] Jung In Park, Mahyar Abbasian, Iman Azimi, Dawn Bounds, Angela Jun, Jaesu Han, Robert McCarron, Jessica Borelli, Jia Li, Mona Mahmoudi, Carmen Wiedenhoef, and Amir Rahmani. 2024. Building Trust in Mental Health Chatbots: Safety Metrics and LLM-Based Evaluation Tools. doi:10.48550/arXiv.2408.04650 arXiv:2408.04650 [cs].
- [40] Michaela C. Pascoe, Sarah E. Hetrick, and Alexandra G. Parker. 2020. The impact of stress on students in secondary school and higher education. *International Journal of Adolescence and Youth* 25, 1 (Dec. 2020), 104–112. doi:10.1080/02673843.2019.1596823 Publisher: Routledge _eprint: <https://doi.org/10.1080/02673843.2019.1596823>
- [41] Hongzhi Qi, Qing Zhao, Jianqiang Li, Changwei Song, Wei Zhai, Dan Luo, Shuo Liu, Yi Jing Yu, Fan Wang, Huijing Zou, Bing Xiang Yang, and Guanghui Fu. 2024. Supervised Learning and Large Language Model Benchmarks on Mental Health Datasets: Cognitive Distortions and Suicidal Risks in Chinese Social Media. doi:10.48550/arXiv.2309.03564 arXiv:2309.03564 [cs].
- [42] K. Jayasankara Reddy, Karishma Rajan Menon, and Anjana Thattil. 2018. Academic Stress and its Sources Among University Students. *Biomedical and Pharmacology Journal* 11, 1 (March 2018), 531–537. <https://biomedpharmajournal.org/vol11no1/academic-stress-and-its-sources-among-university-students/>
- [43] Anita Schick, Jasper Feine, Stefan Morana, Alexander Maedche, and Ulrich Reinighaus. 2022. Validity of Chatbot Use for Mental Health Assessment: Experimental Study. *JMIR mHealth and uHealth* 10, 10 (Oct. 2022), e28082. doi:10.2196/28082 Company: JMIR mHealth and uHealth Distributor: JMIR mHealth and uHealth Institution: JMIR mHealth and uHealth Label: JMIR mHealth and uHealth Publisher: JMIR Publications Inc., Toronto, Canada.
- [44] Christine Schillings, Echo Meißner, Benjamin Erb, Eileen Bendig, Dana Schultchen, and Olga Pollatos. 2024. Effects of a Chatbot-Based Intervention on Stress and Health-Related Parameters in a Stressed Sample: Randomized Controlled Trial. *JMIR Mental Health* 11, 1 (May 2024), e50454. doi:10.2196/50454 Company: JMIR Mental Health Distributor: JMIR Mental Health Institution: JMIR Mental Health Label: JMIR Mental Health Publisher: JMIR Publications Inc., Toronto, Canada.
- [45] Kyoungwon Seo, Samuel Dodson, Negar M. Harandi, Nathan Roberson, Sidney Fels, and Ido Roll. 2021. Active learning with online video: The impact of learning context on engagement. *Computers & Education* 165 (May 2021), 104132. doi:10.1016/j.compedu.2021.104132
- [46] Kyoungwon Seo, Joice Tang, Ido Roll, Sidney Fels, and Dongwook Yoon. 2021. The impact of artificial intelligence on learner–instructor interaction in online learning. *International Journal of Educational Technology in Higher Education* 18, 1 (Oct. 2021), 54. doi:10.1186/s41239-021-00292-9
- [47] Munyi Shea, Y. Joel Wong, Kimmy K. Nguyen, and Patricia D. Gonzalez. 2019. College students' barriers to seeking mental health counseling: Scale development and psychometric evaluation. *Journal of Counseling Psychology* 66, 5 (2019), 626–639. doi:10.1037/cou0000356 Place: US Publisher: American Psychological Association.
- [48] Dominic Ethan Sia, Marco Jalen Yu, Justine Leo Daliva, Jaycee Montenegro, and Ethel Ong. 2021. Investigating the Acceptability and Perceived Effectiveness of a Chatbot in Helping Students Assess their Well-being. In *Proceedings of the Asian CHI Symposium 2021 (Asian CHI '21)*. Association for Computing Machinery, New York, NY, USA, 34–40. doi:10.1145/3429360.3468177
- [49] Thomas Steare, Carolina Gutiérrez Muñoz, Alice Sullivan, and Gemma Lewis. 2023. The association between academic pressure and adolescent mental health problems: A systematic review. *Journal of Affective Disorders* 339 (Oct. 2023), 302–317. doi:10.1016/j.jad.2023.07.028
- [50] Patrick Toulme, Jude Nanaw, and Panagiotis Apostolellis. 2023. Marcus: A Chatbot for Depression Screening Based on the PHQ-9 Assessment. (2023).
- [51] Jenny van Doorn, Katherine N. Lemon, Vikas Mittal, Stephan Nass, Doreen Pick, Peter Pirner, and Peter C. Verhoef. 2010. Customer Engagement Behavior: Theoretical Foundations and Research Directions. *Journal of Service Research* 13, 3 (Aug. 2010), 253–266. doi:10.1177/1094670510375599 Publisher: SAGE Publications Inc.
- [52] Ziang Xiao, Michelle X. Zhou, Q. Vera Liao, Gloria Mark, Changyan Chi, Wenxi Chen, and Huahai Yang. 2020. Tell Me About Yourself: Using an AI-Powered Chatbot to Conduct Conversational Surveys with Open-ended Questions. *ACM Trans. Comput.-Hum. Interact.* 27, 3 (June 2020), 15:1–15:37. doi:10.1145/3381804
- [53] Xuhai Xu, Bingsheng Yao, Yuanzhe Dong, Saadia Gabriel, Hong Yu, James Hendler, Marzyeh Ghassemi, Anind K. Dey, and Dakuo Wang. 2024. Mental-LLM: Leveraging Large Language Models for Mental Health Prediction via Online Text Data. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 1 (March 2024), 31:1–31:32. doi:10.1145/3643540
- [54] Muhammad Saiful Bahri Yusoff, Liew Yee, Ling Wei, Loke Meng, Lim Bin, Tan Siong, and Ahmad Fuad. 2011. A study on stress, stressors and coping strategies among Malaysian medical students. *International Journal of Students' Research; Vol 1, No 2 (2011): Taking Research Forward; 45-50* 1 (April 2011). doi:10.5549/IJSR.1.2.45-50
- [55] Sharon Ziv-Beiman. 2013. Therapist self-disclosure as an integrative intervention. *Journal of Psychotherapy Integration* 23, 1 (2013), 59–74. doi:10.1037/a0031783 Place: US Publisher: Educational Publishing Foundation.

A SISCO Inventory of Academic Stress (SISCO-AS)

Thank you for participating in the SISCO Inventory of Academic Stress (SISCO-AS) questionnaire. Your responses are invaluable in helping us better understand academic stress. Please respond using

a 5-point Likert scale from 1 to 5, where 1 is ‘never,’ 2 is ‘rarely,’ 3 is ‘sometimes,’ 4 is ‘almost always,’ and 5 is ‘always.’

Stressors: On a scale from 1 (never) to 5 (always), please indicate how often each of the following situations has caused you stress.

- (1) Intense competition
- (2) Assignment overload
- (3) Relationship with the professor
- (4) Evaluation
- (5) Academic workload
- (6) Course difficulty
- (7) Burden of class participation
- (8) Limited time

Symptoms: On a scale from 1 (never) to 5 (always), please indicate how often you experienced each of the following reactions when feeling worried or nervous.

- (9) Sleep disorders
- (10) Chronic fatigue
- (11) Headaches or migraines
- (12) Digestive problems
- (13) Nail-biting
- (14) Drowsiness
- (15) Restlessness
- (16) Feelings of depression
- (17) Anxiety, distress, or desperation
- (18) Concentration problems
- (19) Increased aggression
- (20) Tendency to argue
- (21) Isolation from others
- (22) Lack of motivation
- (23) Increased or decreased food intake

Coping strategies: On a scale from 1 (never) to 5 (always), please indicate how often you used each of the following strategies to manage situations that caused you worry or nervousness.

- (24) Assertive skills
- (25) Developing a plan and executing tasks
- (26) Self-praise
- (27) Religious beliefs
- (28) Seeking information about the situation
- (29) Venting emotions and confiding in others

Thank you for completing the survey. We appreciate your time and thoughtful responses.

B Narrative analysis of participant responses

B.1 Reviewing the experience

Participants in both NSD and SD chatbot conditions shared positive feedback during the ‘reviewing the experience’ stage. Responses are labeled as either from the NSD chatbot condition (‘NSD’) or the SD chatbot condition (‘SD’). In the NSD chatbot condition, participants appreciated the chatbot’s neutral tone, citing a sense of safety and openness. One participant noted it was easier to “reconsider my stress without feeling pressured” (NSD 17), while another described a “safe space to talk, where I wouldn’t be judged” (NSD 3). Several participants also felt the chatbot helped them organize their

thoughts: “answering the chatbot’s questions helped me organize my thoughts about my academic stress” (NSD 6). Another highlighted, “it was helpful to have a calm, objective conversation that didn’t push me to feel a certain way” (NSD 12). Overall, the NSD chatbot created a comfortable, non-judgmental environment for reviewing academic stress.

Participants in the SD chatbot condition praised the chatbot’s empathetic tone and self-disclosure for fostering deeper understanding and more detailed sharing. Several likened the interaction to confiding in a trusted friend. One participant recalled, “it felt like talking to a friend who really understood what I was going through” (SD 19), while another noted that the chatbot’s self-disclosure made me feel less alone, saying, “the way the chatbot shared its own stories made me feel like I wasn’t alone” (SD 16). This empathetic approach heightened awareness of stressors. One participant mentioned, “it helped me realize there were certain stressors I hadn’t noticed before” (SD 8), and another said, “the chatbot’s stories made me think more about my own experiences and feelings” (SD 1). Overall, the SD chatbot facilitated more meaningful reflections on academic stress.

B.2 Critical analysis

Participants in the NSD and SD chatbot conditions approached the ‘critical analysis’ stage differently, with notable differences in reflection depth. In the NSD chatbot condition, participants described the interaction as rigid and formulaic, limiting deeper stress analysis. Many provided surface-level responses and found that repetitive questions hindered thoughtful exploration. One participant recalled, “the chatbot made me think about my stress, but it felt like I was just answering the same questions over and over” (NSD 7), while another said, “I responded quickly, like I was filling out a form” (NSD 15). This rigidity often prevented meaningful connections to personal experiences, with some calling the questions “intrusive yet unproductive” (NSD 4) or “one-sided” (NSD 19). Consequently, interactions in the NSD chatbot condition remained superficial, with limited engagement in deeper self-reflection.

In contrast, participants in the SD chatbot condition reported that the chatbot’s self-disclosure and relatable questions encouraged deeper inquiry, helping them connect experiences to broader emotional and cognitive frameworks. The SD chatbot’s relatable examples and prompts made participants more comfortable exploring stressors and emotions. One participant explained, “the chatbot’s stories made me think about my own situation in a new way” (SD 6), prompting deeper questions such as, “How did I feel at that moment?” and “Why did I react that way?” Participants also noted that the SD chatbot helped them contextualize reflections within their personal values. For instance, one participant said, “the chatbot’s examples helped me realize how I react to stress and what I could do differently” (SD 5), while another shared, “hearing relatable stories made me reflect more deeply on how I handle stress” (SD 20). Overall, the SD chatbot encouraged a more insightful reflective process, enhancing participants’ understanding of their academic stress.

B.3 Reflective outcome

Participants in the NSD and SD chatbot conditions showed notable differences in reaching the ‘reflective outcome’ stage, where reflections are translated into actionable strategies. In the NSD chatbot condition, reflections often reinforced existing thoughts without offering new insights. For instance, one participant noted, “it made me think about my stress, but I didn’t realize anything new” (NSD 1), and another said, “the interaction didn’t help me understand my stress any better” (NSD 10). The chatbot’s structured approach limited deeper insights and hindered the application of reflections to practical coping strategies. While it helped articulate stress, participants felt it offered little guidance for managing it. As one explained, “the chatbot helped me describe my stress, but it didn’t suggest anything I could actually do” (NSD 13), and another admitted, “I’m still unsure how to effectively manage my stress” (NSD 20). Consequently, interactions often remained focused on identifying stressors without progressing to actionable planning.

In contrast, participants in the SD chatbot condition consistently reported gaining new insights and identifying practical ways to manage their stress. The chatbot’s detailed questions and reflective guidance encouraged exploration of previously overlooked causes and new coping strategies. One participant shared, “the chatbot asked specific questions that made me realize there were underlying causes I hadn’t noticed before” (SD 12), while another noted, “Its reflective prompts helped me identify effective coping strategies that I hadn’t considered previously” (SD 10). Participants also reported success translating reflections into practice. One explained, “discussing specific situations helped me understand my usual coping strategies and how to use them better” (SD 7), while another remarked, “I realized there are more ways to manage stress than I thought, and I want to try some of them” (SD 18). These findings suggest that the SD chatbot fostered deeper reflection and helped participants develop actionable plans for more effective stress management.